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Conflict-Resolution Heuristics for En Route Air Traffic Management

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This paper describes the first study in a PhD program examining the effect of mental workload on conflict resolution decisions in Air Traffic Control. This study focussed on the heuristics controllers use when resolving potential conflicts between aircraft and highlighted contextual factors that affect perceived workload. Using static maps of novel scenarios, controllers were interviewed about how they would resolve potential conflicts and what factors contributed to their workload. Controllers reported that they scanned in a clockwise and top-bottom pattern; grouped aircraft with similar characteristics and used a range of different lateral and vertical conflict resolution heuristics. This research expands on previous conflict resolution heuristics and identifies the sources of complexity, which influence decision choices.

Whilst a great deal is known about the way that air traffic controllers detect conflicts amongst aircraft (e.g., Loft, Bolland, Humphreys & Neal, 2009), much less is known about the way that they resolve conflicts. Conflict resolution is a complex and demanding task, often performed under time pressure. As human controllers have a limited cognitive capacity, they are unable to comprehensively search the solution space to find an optimal solution. In these environments, simple heuristics can be used to make fast decisions with little information (Hutchinson & Gigerenzer, 2005). These simple heuristics work surprisingly well and are what humans widely use in a range of different environments. The aim of this study was to identify some of the heuristics that controllers use when resolving potential conflicts, and the sources of complexity that may influence the choice of strategy.

Controllers often report that they have the equivalent of a "library" of solutions to particular problems (Kallus, Van Damme, & Dittman, 1999). When a conflict is detected, controllers access their library to find a previous solution and then adapt and apply that previous solution to the new case. If no appropriate solution is found, controllers either review their solution library for a suitable plan or use problem-solving techniques to develop a new solution.

In addition, D'Arcy and Della Rocco (2001) argue that controllers change the way they work when they become busy. They become more conservative, act earlier, plan more and communicate to minimise repeats. Sperandio (1978) notes that as workload increases, controllers do not passively accept the increase in the number of aircraft. Instead, they adapt their heuristics to deal with greater workload. Sperandio (1978) argued that controllers have a pool of strategies from which they sample and as workload increases, the more economic ones tend to be sampled more often.

In terms of extracting conflict resolution heuristics, several qualitative methods have been used by Callantine (2002) and Kirwan and Flynn (2001). Both describe each solution type sequentially and in Kirwan and Flynn's (2001) study, the researchers identified formal (e.g., speed regulations) and informal (e.g., parallel headings) rules, general principles (e.g., don't rely on climb performance above FL200) and the main factors influencing conflict resolution choice (e.g., destination). These formative studies however, did not identify the actual resolutions controllers used for each conflict in each scenario or the degree of agreement for such solutions.

In summary, prior research has begun to suggest that controllers have a set of heuristics that they have acquired by experience, and that the selection of heuristics may be determined by complexity. In the present research, we used a think-aloud procedure to identify what those heuristics may be. Controllers were presented with static screen shots of novel scenarios and given sufficient time to discuss their traffic management plan. A qualitative analysis was conducted to provide a broad overview of heuristics used across different scenarios, contexts and individual controllers.

METHOD

Participants

Fourteen male, En route radar endorsed ATC operators from Brisbane Centre participated in this study. The biased gender distribution towards male participants accurately reflects the proportions typically observed in this domain. Controllers held endorsements across both radar and procedural sectors.

Materials

Five static screen shots of traffic, which varied in terms of complexity, were developed by subject matter experts (SMEs) at Melbourne Centre. SMEs were instructed to develop scenarios with low traffic complexity (scenarios one and two) and scenarios with high traffic complexity (scenarios three, four and five). Scenario maps used generic backgrounds each containing six route structures (e.g., H66) and two holding patterns (on Q181 and H66). All routes were treated as both inbound and outbound. Each scenario had between five (scenario one) and 10 (scenario six) aircraft. The following aircraft types were used: 1124 Westwind (WW24), Canadair Challenger (CL60), Cessna 550 Citation II (C550), Boeing 777-200 (B777), Boeing 737-800 (B738) and the Boeing 747-100/200 Freighter (B74F). Callsigns contained three-letter designators (e.g., TTO). Both familiar aircraft designators (e.g., QFA for a QANTAS aircraft) and unfamiliar aircraft designators (e.g., SSE) were used. Aircraft were directed along the major routes in both directions. Faster following, crossing tracks and opposite direction conflicts

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were scripted. Scenarios contained between two (scenario one) and 28 (scenario three) potential pairwise conflicts. Screen shots were printed in colour on A2 sheets and laminated.

Procedure

Subjects were presented with static maps of novel scenarios (the incidents) and asked to provide verbal protocols. Discussing traffic management heuristics from visual aids removed potential memory fallacies from recalling past situations. Open-ended probe questions were used to identify the decision processes controllers used when managing traffic. Informed consent was collected and a brief handover was given before each scenario was presented. For example, controllers were told the following: *"This is a fictitious sector in En Route, radar control. The major routes are H66, Q181, W421 and V76. There are eight aircraft under your jurisdiction although movement has been frozen. Weather conditions and coordination do not need to be taken into account and there is nil wind."*.

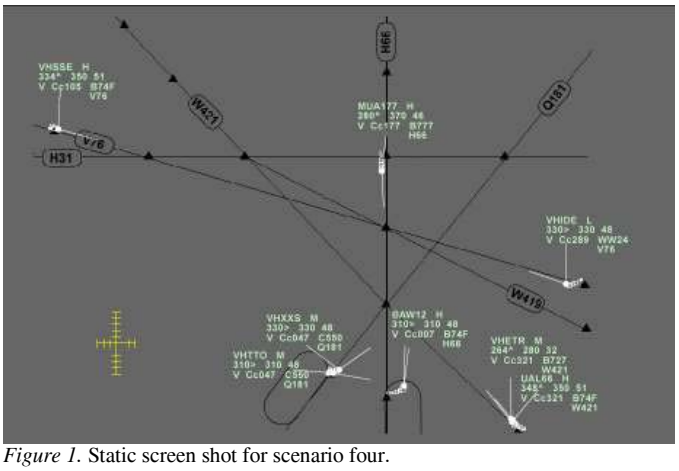


Figure 1. Static screen shot for scenario four.

Participants were asked to observe the scenario and to describe what they would do if they were controlling the sector. They received the following instruction: *"Please go step by step as you work through this scenario and, as much as possible, explain what you are thinking about. As you scan please tell us what you are looking at."*

Scenarios were presented in random order. Controllers spent approximately 20 minutes verbalising their traffic management plan of each scenario. They were asked to talk about all their actions including those where they might search for information, evaluate potential problems or issues, project ahead, consider options and decide on actions or priorities. Participants were also asked to identify the factors that contributed to the complexity of each scenario. Interviews were videotaped and the data recorded.

RESULTS

The analyst coded the aircraft that the controllers focused on, the actions they indicated that they would take to resolve the conflicts in each scenario, together with their considerations at each stage. The coding provided a sequential break-down of the steps that each participant would take to solve each scenario. The heuristics that controllers used are described below, together with illustrative comments from controllers.

Scanning and Conflict Detection

The general scanning method controllers employed was in a clockwise and top-to-bottom direction around the screen. Controllers stated that they generally used this scanning pattern across all scenarios, unless critical information drew their focus to a key area. For example, controller 13 stated: "*"I look and work left to right, clockwise. In normal airspace I look top to bottom or if not, bottom to top depending on where major aerodromes are.*". Across all scenarios controllers reported that they made between three and six scans or that they scanned in five to ten second intervals. Controller 14 commented: *"I scan three or four times…look at each aircraft before the picture is in my head."*.

Controllers applied three main conflict detection heuristics. The first heuristic involved *checking levels, tracks and headings.* Controllers reported that they primarily focussed on these aspects and typically checked levels first. Controller 12 stated: *"I check for levels that are in conflict and then which way aircraft are going. Get rough idea where aircraft are, then see levels and where they are going."*. The second conflict detection heuristic was *prioritisation of aircraft.* Controllers used critical information to prioritise the order in which to attend to aircraft. Aircraft were typically considered a high priority if they were in close proximity, converging and/or on climb. Controller two stated: *"Aircraft with the closest proximity are looked at first, so I am looking at the levels of VHTTO and VHHRD first. Then my personal judgement is that the next two that were going to come into conflict were VHCCC and QFA22."*. The third conflict detection heuristic was *grouping aircraft.* Controllers grouped aircraft together if they had similar characteristics, if they were in potential conflict with each other or if they were in close proximity. Controller 13 stated: *"I group close proximity aircraft first, the bottom cluster, then work outside that."*.

Conflict Resolution Heuristics

Controllers relied on five lateral (vectoring) heuristics and eight vertical (level change) heuristics to resolve potential conflicts between aircraft. As expected, the complexity of the airspace appeared to influence heuristic preference. Level changes were more commonly preferred to other solution types (e.g., vectors, speed changes) under high workload conditions. With more time available, controllers could assign the highest vacant level a climbing aircraft could reach, in order to minimise disruption to an aircraft's climb profile and to keep the aircraft on its planned trajectory. With less time available, controllers were more likely to issue level changes closer to the aircraft's current level, as reaching separated levels sooner resolves the potential conflict more expediently. For example, controller five commented: *"You could vector them but it is a bit congested at the moment, so I am more inclined to give an immediate level change"*. Under extremely high workload demands however, controllers' preferences favoured reactive, lateral (vectoring) solutions. For example,

controller two stated: *"Under extreme time pressure I will favour vectors if aircraft cannot descend, as it gives separation assurance the quickest. In cases where they couldn't drop down (descend) I went for the vector. No point asking aircraft to climb if they come back and say they can't".*

The 13 (five lateral, eight vertical) conflict resolution heuristics are described below with qualitative comments.

1. Point behind other aircraft. This heuristic involves vectoring one aircraft either directly at or just behind a potentially conflicting aircraft. It requires a controller to issue one heading change instruction to an aircraft, listen for the read-back and then monitor the aircraft's altered trajectory. It is considered a 'set and forget' solution, as it only requires one instruction. It can be used to assure separation quickly, in order to reduce future workload. This heuristic was employed by several controllers in scenario one. In this scenario there is a potential conflict between QFA137 and BAW12. These aircraft are on crossing tracks and both are maintaining FL350. The solution is to vector QFA137 to pass behind BAW12. A comment from controller two who implemented this heuristic was: *"Give him (QFA137) a turn to miss BAW12. No point asking him (QFA137) to climb if he comes back and says he can't. Vector would give separation assurance quickest."*. These aircraft are presented in Figure 2.

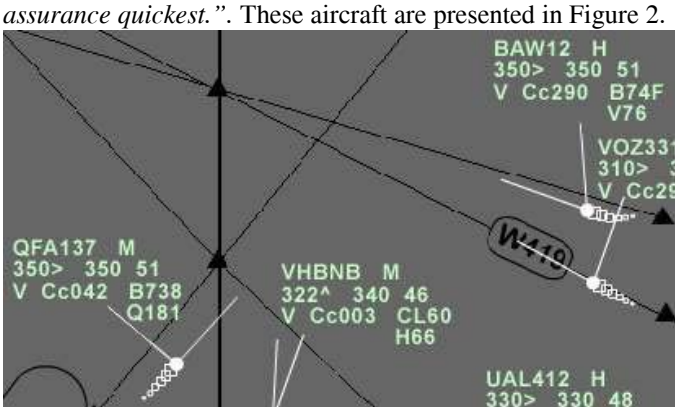


Figure 2. Potential conflict between QFA137 and BAW12.

2. Direct away from potential conflicts ("Take out"). This heuristic involves assigning an aircraft a short trajectory deviation from its planned route. Following this new trajectory (vector) will ensure that the aircraft does not breach five miles separation with other aircraft occupying the same level. It requires a controller to issue a heading change, listen for the readback and monitor the new trajectory. This solution can be used to assure separation very quickly, as it only requires one instruction. A controller however, will create future workload with this heuristic, as they have to monitor the new trajectory and redirect the aircraft back to its original route in the future. A comment from controller three who implemented this heuristic was: *"It is too late to try and change VHTTO's level. BAW12 is going north and is heavy. VHTTO is a medium jet, so must turn him away cause can't change his level quickly".*

3. Parallel Track. This heuristic involves assigning an aircraft a new trajectory, which is parallel to its current route. Using this trajectory for a short time, ensures that the aircraft can remain separated (by five miles) from other aircraft occupying the same level. It requires a controller to issue a heading change, listen for the readback and monitor the new trajectory. This solution can be used to assure separation very quickly, as it only requires one instruction. Again however, this will increase a controller's later workload, as the aircraft's new trajectory needs to be monitored and the aircraft will eventually need to be redirected back to its original route. A comment from controller three who implemented this heuristic was: *"I am taking VHGTR out of the scenario straight away because he is too close (to VHCCC). VHGTR will go south. Paralleling will get him (VHGTR) away from VHCCC.".*

4. Take out for five miles, then put back on track. This heuristic involves assigning an aircraft a new trajectory, which will keep the aircraft five miles either right or left of its current route. Using this trajectory for a short time, ensures that the aircraft remains separated (by five miles) from other aircraft occupying the same level. It requires a controller to issue a heading change, listen for the readback and monitor the new trajectory. This solution can be used to assure separation very quickly, as it only requires one instruction. Again however, this increases a controller's future workload, as the aircraft's new trajectory needs to be carefully monitored and the aircraft needs to be directed back to its original route in the near future. A comment from controller nine who implemented this heuristic was: *"Bit late to chop VHGBT off now as he is already out of 320. Only option would be vector. A vector off right points (VHGBT) away from conflicting aircraft. I will take him off track to get 5 miles, then put back on track.".*

5. Pass in front. This heuristic involves assigning an aircraft a trajectory, which ensures it will pass in front of a potentially conflicting aircraft by at least five miles. It requires a controller to estimate the approximate crossing points and crossing times of the aircraft (based on the current performance characteristics of the aircraft), issue the vector, listen for the readback, monitor the new trajectory and eventually redirect the aircraft back to its original route. This solution can be used to assure separation, although the solution is very time consuming to calculate and monitor. A comment from controller five who implemented this heuristic was: *"I will point (UAL66) to go in front of VHXXS' vector. Heading Forth 350 degrees.".*

6. Cut off at nearest available level on climb. This heuristic involves amending an aircraft's cleared flight level to the nearest available level on its climb. It requires a controller to issue a level change instruction to an aircraft, listen for the read-back and then monitor the aircraft's climb to the amended level. This solution can be used to assure separation quickly, as the aircraft will reach the new level faster than the previous cleared flight level. It requires less visual monitoring by the controller and only requires one instruction. This heuristic was employed by several controllers in scenario four. In this scenario, MUA177 is in potential conflict with VHTTO, VHXXS, BAW12 and UAL66. MUA177 (currently at FL280) is heading in an opposite direction to these aircraft and is on climb through all of their levels (to FL370). The solution is to amend MUA177's cleared level to the nearest level on its climb (FL290). This level will be at least 1000 feet below the levels of all conflicting aircraft by the crossing point. A comment from controller 10 who implemented this heuristic was: *"MUA177 is climbing through the pack so need*

to eliminate. I considered 280 but no good due to VHETR and MUA177 already at 280. So will give 290 for four minutes until he (MUA177) comes out the other side of the pack..". These aircraft are presented in Figure 3.

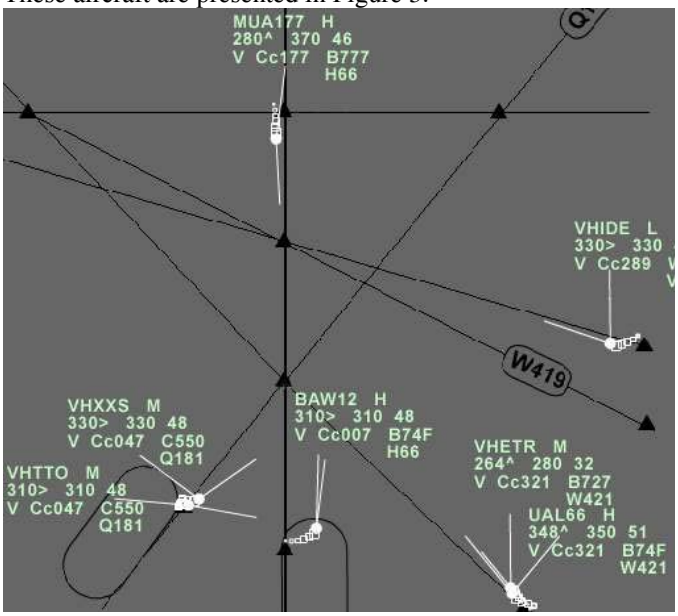


Figure 3. Potential conflict between MUA177 and VHTTO, VHXXS, BAW12, VHETR, UAL66 and VHIDE.

7. Cut off at highest possible level on climb. This heuristic involves assigning a new cleared flight level to an aircraft, which is the highest vacant level the aircraft can reach by the crossing point with the potentially conflicting aircraft. It requires a controller to estimate how high the aircraft can climb by the crossing point based on its current performance, assess whether the level is vacant, issue the new flight level, listen for the readback and monitor the climb. This solution can increase a controller's workload, as they need to calculate which level is the highest that can be reached and monitor the climb to ensure the aircraft levels off at the amended cleared flight level. A comment from controller 12 who implemented this heuristic was: *"Looking at aircraft types, would have to chop (VHTRE) off underneath VHEWR.I have got one minute short route probes. They will be past each other in five minutes, then I can put people up to the levels they want."*

8. Request nearest level above conflicting aircraft. This heuristic involves assigning a new cleared flight level to an aircraft, which is the next highest level (in intervals of 1000ft) above the potentially conflicting aircraft. It requires a controller to determine whether the aircraft can sufficiently reach this level by the crossing point, issue the new level, listen for the readback and monitor the climb to the new level. This is often not the fastest solution, as the controller needs to calculate whether the aircraft can reach this level in time and subsequently monitor the climb. A comment from controller one who implemented this heuristic was: *"I would highlight VHTTO and give it 310. This solves it straight away. I would cut (VHTTO) off at the nearest level on its descent that is above VHERD."*.

9. Descend to nearest available level. This heuristic involves assigning an aircraft the nearest available level (in intervals of 1000ft). It requires a controller to assess whether the level is going to be free, issue the level change, listen for the readback and monitor the decent. While this solution can be used to assure separation very quickly, as the controller only has to issue one instruction and it is a short descent, it poses a penalty for the aircraft. By descending to a lower level, the aircraft will use more fuel than it would at higher altitudes. It also increases a controller's future workload, as they may need to re-climb the aircraft back up to or through the level it was originally descending from. It is not usually preferred and is primarily used in periods of high workload when other options are not available. Controller one for example referred to this as a 'ding ding' heuristic which he uses in immediate situations when two aircraft are at the same level on converging tracks. A comment from controller two who implemented this heuristic was: *"Turning (VHTTO) is not really an option because they (BAW12 and VHTTO) would get too close. Could have given a left turn for VHTTO but am going for immediate descent. Would give him (VHTTO) 300."*

10. Assign the only level available (ensure every aircraft is maintaining a different level). This heuristic involves amending an aircraft's cleared flight level to the only vacant level available. It requires a controller to assess which level is vacant and whether the aircraft can reach this separated level by the crossing point, issue the instruction and monitor the aircraft to the new level. This may not be an expedient solution, as the level may not be near to the aircraft's current altitude. This heuristic is typically used in busy periods to keep all aircraft separated at different levels. A comment from controller five who implemented this heuristic was: *"I think VHEWR will make that level (270) before getting involved with all of the central aircraft. It might infringe in 90 seconds time."*.

11. Step climb/descent. This heuristic involves incrementally amending an aircraft's cleared flight level in intervals of 1000 feet to 'step' it past the potentially conflicting aircraft using vertically separated levels. It requires a controller to issue a number of new cleared flight levels to the aircraft, listen for readbacks and monitor the climb/descent actions to each new level. This solution may be used to assure separation quickly, although the complete solution is intensive on a controller's future workload. Each new level requires focused attention on monitoring and listening to readbacks. It does however, allow controllers to keep aircraft on their descent/climb without interrupting their trajectories or restricting them at certain levels for long periods. A comment from controller nine who implemented this heuristic was: *"Looking at their directions one of them could change. VHEWR is already a problem so would give him a lower level. I would step climb VHEWR above VHTRE. Give VHEWR FL310 to keep under VHTRE."*.

12. Expedite climb/descent. This heuristic involves issuing an instruction to an aircraft to expedite their climb/ descent to their cleared flight level. It requires a controller to issue the instruction, listen for the readback and monitor the climb/descent. This solution can be used to assure separation quickly, as the aircraft will now reach their cleared level faster. A comment from controller three who implemented this heuristic was: *Will still have in excess of five nautical miles based on the rough scale but will change BFB's rate of climb*

reference UAL412 because don't have 1000 feet with him. Will give a hurry-up requirement.".

13. Report maintaining (to ensure aircraft are at least 1000 feet apart). This heuristic serves as a check that aircraft will remain separated by the relevant standards. It requires a controller to issue a request to the aircraft to report back to them when they reach a particular level. A comment from controller five who implemented this heuristic was: *"Looking at directions QFA22 will be a real problem with VHEWR. Checking speeds and their meeting point I expect QFA22 to get to 350 very quickly and will ask him to report maintaining and confirm his present heading.".*

Sources of Complexity

Controllers stated that different factors contributed to complexity in high compared to low workload conditions. In high workload scenarios controllers identified the following sources of complexity: immediate concerns, many aircraft changing levels in a small space, aircraft descending/climbing through another, the number of conflicts, difficulty seeing changing levels, time constraints, aircraft being off track, aircraft type, communication, aircraft in close proximity, narrow focussing, aircraft pointing at each other and converging, familiarity with the airspace, vector monitoring, having to keep a lot in (working) memory, moving labels around, having more to consider and having more options. In low workload scenarios controllers identified fewer sources of complexity. These included: aircraft descending/ climbing through each other, aircraft being off track, aircraft pointing at each other and converging, moving labels around, the difficulty of seeing changing levels, aircraft not maintaining levels, narrow focussing, aircraft type, communication and familiarity with the airspace.

DISCUSSION

The results provide an understanding of the heuristics controllers use for scanning, conflict detection, and conflict resolution. Scanning was conducted in a clockwise and top-bottom pattern, in intervals of five to ten seconds. Building on previous conflict detection research (Seamster, Redding, Cannon, Ryder & Purcell, 1993; Endsley & Smolensky, 1998; Nunes & Mogford, 2003), controllers made judgements, in order to identify a conflict by considering altitude, location and route. Controllers used three primary conflict detection heuristics: focusing on levels, tracks and headings; prioritising key aircraft and grouping aircraft with similar characteristics.

In terms of conflict resolution, as expected controllers used a series of lateral and vertical heuristics to keep aircraft separated. Building on previous work (Kirwan & Flynn, 2001; Callantine, 2002), a series of solution types were observed. Some of these were consistent with previously identified rules (e.g., 'parallel headings' is similar to the 'parallel track' heuristic) and principles (e.g., 'Turn slower aircraft behind' may be similar to the 'point behind' heuristic). Expanding on earlier research however, this study identified the actual resolutions controllers used for each conflict in each scenario, and some degree of agreement for these solutions. Furthermore, in the present study the list of heuristics included a broader range of alternatives, which included altitude and climb/descent solutions (e.g., step descent/climb), as well as spacing and general separation rules.

Complexity also appeared to influence heuristic choice. As D'Arcy and Della Rocco (2001) and Sperandio (1978) argued, as workload increased, controllers stated that they adapted their heuristics to deal with greater complexity. The sources of complexity identified were similar to those previously identified in the literature (e.g., Hughes, 2000).

There are limitations associated with the methodology used in the current study. Specifically, the information presented is subjective in nature and there may well be a discrepancy between what controllers say they do, and what they actually do. Furthermore, while these data are useful for identifying the types of heuristics that controllers may use in generic situations, more work is needed to understand the constraints that influence heuristic selection. To overcome this, future experimental studies examining conflict resolution methods under different levels of workload were designed within this research program.

Practically, these resolution heuristics can be incorporated into training programs, rostering decisions and sector design. The research also has specific applications towards the design and development of future automated decision making tools (e.g., conflict resolution algorithms) to facilitate controller performance. With further investigation, these criteria could be incorporated into algorithms, which are widely accepted by controllers.

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